

Brandon Moy

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EDUCATION

Stony Brook University - Honors College

Expected Graduation: May 2028

B.S. in Computer Science and Applied Mathematics & Statistics

GPA: 3.79/4.0

Relevant Coursework: Systems II, Software Development, Data Structures, Analysis of Algorithms

EXPERIENCE

Software Engineering Intern

May 2025 - Aug 2025

Get Talky

- Built full-stack features for an AI conversation evaluation platform using React, Node.js, and PostgreSQL.
- Designed **REST APIs** and PostgreSQL schemas for comment tracking and review status, with integration tests.
- Built client and global-level **CSV exports** of comment data, streamlining reporting for all clients.
- Refactored React comment state to update locally instead of **re-fetching** the full thread on each post.
- Prototyped UI automation with AskUI for client testing, cutting manual QA validation time by **20%**.

Undergraduate Researcher

Jun 2026 - Present

Stony Brook University - Algorithms Research, Prof. Pramod Ganapathi

- Derive and prove correct **dynamic programming** algorithms for an algorithmic problem-solving book.
- Optimized multi-peg Tower of Hanoi DP from $O(kn^2)$ to $O(nk)$ via convexity and monotone split points.
- Constructed a **counterexample** to a 2025 arXiv paper's split-point formula, verified by exhaustive BFS.
- Cut memory $O(n^2)$ to $O(n)$ with rolling arrays and derived wavefront-parallel loop orders for DP tables.

Undergraduate Teaching Assistant

Jan 2026 - May 2026

Stony Brook University - Object-Oriented Programming

- Led weekly Java labs for **100+ students** on object-oriented design, inheritance, and debugging strategies.
- Live-coded lab solutions in front of students and debugged their projects during office hours.

PROJECTS

Real-time Fraud Decisioning Platform | *Python, FastAPI, PostgreSQL, Redis, Kafka, Docker* [GitHub](#)

- Built a real-time fraud-scoring API with a Redis feature store and Kafka pipeline across **10 Docker services**.
- Raised throughput from ~200 to **500 tx/s** with zero errors after k6 tests exposed a single-process GIL bottleneck.
- Designed **atomic Redis Lua** sliding windows so concurrent requests can't bypass velocity limits.
- Ensured **at-least-once** Kafka delivery via a transactional outbox, DLQ, and idempotent consumers.

Undergrad Course Planner | *React, D3.js, Node.js, Express, Supabase, OpenAI*

[GitHub](#) | [Demo](#)

- Built a course-planning platform that visualizes prerequisite graphs and degree progress for **100+ students**.
- Developed a **Node.js/Express** backend on Supabase (PostgreSQL) serving course data and student progress.
- Built an AI chatbot summarizing scraped RateMyProfessors reviews with Selenium and OpenAI.

MP3 Metadata & Audio Tool | *C, Make, Criterion*

- Built a C command-line tool for **byte-level parsing** of ID3 tags and MPEG frame headers.
- Implemented PCM trim and overlay across mismatched sample rates and mono/stereo layouts.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js, Express, React, D3.js, pytest, pandas, NumPy, scikit-learn

Backend & Infrastructure: PostgreSQL, Redis, Kafka, Docker, Linux, GitHub Actions, Prometheus, Claude Code

Concepts: Distributed Systems, Concurrency, Event-Driven Architecture, Message Queues

ACTIVITIES

Gaming - Fortnite: Top 10 in official Epic Games tournament (\$500 prize value); **Valorant:** Immortal (~1% NA)

Stony Brook Computing Society - Representative

Competitive Club Teams - Rock Climbing and Volleyball